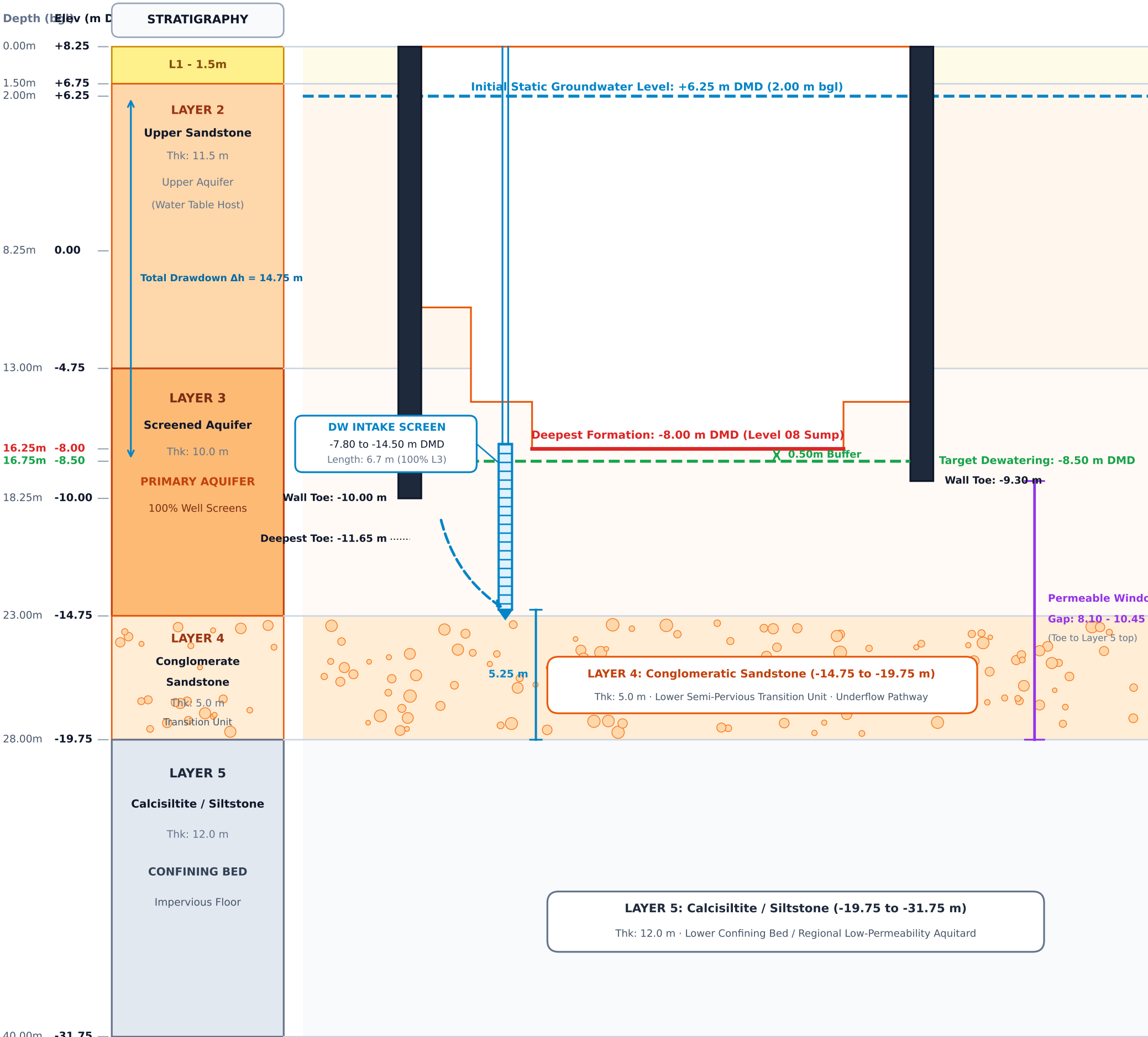


VERTICAL CROSS-SECTION: HYDROSTRATIGRAPHY, EXCAVATION & DEWATERING (5 STRATA)

Datum: Dubai Mining Datum (DMD) · Baseline GWL: +6.25 m · Deepest Formation: -8.00 m · Target Dewatering: -8.50 m



VERTICAL SECTION LEGEND

1. HYDROSTRATIGRAPHIC STRATA (5 LAYERS)

- Layer 1: Dune Sand / Silt (1.5 m)**
+8.25 to +6.75 m DMD · Vadose Zone
- Layer 2: Upper Sandstone (11.5 m)**
+6.75 to -4.75 m DMD · Upper Aquifer / GWL
- Layer 3: Screened Aquifer (10.0 m)**
-4.75 to -14.75 m DMD · Kh: 4.32 m/d · Sy: 0.12
- Layer 4: Conglomerate Sandstone (5.0 m)**
-14.75 to -19.75 m DMD · Transition Unit
- Layer 5: Basal Confining Bed (12.0 m)**
-19.75 to -31.75 m DMD · Regional Aquitard

2. ENGINEERING & DEWATERING

- Initial Static GWL: +6.25 m DMD**
Depth: 2.00 m bgl (approved baseline)
- Deepest Excavation: -8.00 m DMD**
Level 08 elevator & sump pits (CONFLICT-002)
- Target Dewatering: -8.50 m DMD**
0.50 m dry buffer below deepest formation
- Secant Pile Cutoff Wall (168 m)**
Toes at -9.30, -10.00, -11.65 m DMD
- Deepwell Intake Screen (6.7 m)**
-7.80 to -14.50 m DMD (wholly in Layer 3)

3. HYDROGEOLOGICAL DYNAMICS

- Underflow Seepage Pathway**
Upward flow beneath partially penetrating toe
- Permeable Window: 8.10 - 10.45 m**
Vertical gap between toe and Layer 5 top
- Well Toe Clearance: 5.25 m**
Distance from well toe (-14.50m) to Layer 5 top

SYSTEM ABSTRACTION DISCHARGE:

- Steady Total: 46.5 m³/hr (1116 m³/day)
- Per Well: 7.75 m³/hr (6 active deepwells)
- Peak Total: 93.0 m³/hr (15.5 m³/hr/well)

KEY HYDROGEOLOGICAL FINDINGS:

- Secant pile wall terminates inside Layer 3 at -9.30 to -11.65 m DMD.
- Shoring does NOT penetrate Layer 5 basal confining bed (-19.75 m).
- A 8.10 to 10.45 m open permeable gap exists beneath wall toe.
- 100% of deepwell intake screens (6.7 m) are in Layer 3.
- Well toe terminates at -14.50 m DMD (5.25 m clearance to L5).
- Layer 4 (conglomerate, 5.0 m) provides deep transition pathway.
- Layer 5 acts as effective impervious base for flow modelling.
- Target dewatering achieves 0.50 m dry buffer below -8.00 m.
- Total operational duration: 150 days (7 d peak + 143 d steady).